

COMPUTER NETWORKS

Unit-2: Error Detection and Correction & Data Communication Media

Comprehensive Study Notes

HPU BCA Semester VI

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2.1 Introduction to Errors in Data Communication

In data communication, **errors** occur when data is corrupted during transmission due to various factors like noise, interference, or signal degradation.

Why Errors Occur

Causes	Description
Noise	Electrical interference in the medium
Attenuation	Signal weakening over distance
Interference	External signals disrupting transmission
Distortion	Signal shape changes during transmission
Crosstalk	Signals from adjacent wires interfering

2.2 Types of Errors

2.2.1 Single-Bit Error

Theory: Only one bit in the data unit is changed from 0 to 1 or 1 to 0.

Example: Original data: **10110010** → Received data: **10110011**

Original: 1 0 1 1 0 0 1 0

Received: 1 0 1 1 0 0 1 1

^ (bit 1 changed to 1)

Characteristics:

- Affects only **one bit**
- **Rare** in modern networks (more common in parallel transmission)
- Occurs due to **white noise**

2.2.2 Burst Error

Theory: Two or more consecutive bits in the data unit are changed.

Example: Original: **10110010** → Received: **11001010**

Original: 1 0 1 1 0 0 1 0

Received: 1 1 0 0 1 0 1 0

^ ^ ^ ^ ^ (burst of 5 bits corrupted)

Characteristics:

- Affects **multiple consecutive bits**
- **Very common** in communication systems
- Caused by **impulse noise**, lightning, power surges

Burst Error Length: First corrupted bit to last corrupted bit; bits in between may or may not be corrupted.

Comparison: Single-Bit vs Burst Error

Feature	Single-Bit Error	Burst Error
Bits Affected	Only one bit	Multiple bits
Frequency	Rare	Common
Cause	White noise	Impulse noise
Detection	Easier	More difficult
Correction	Easier	More complex

2.3 Error Detection

Error detection is the process of identifying whether errors have occurred during transmission.

2.3.1 Redundancy

Extra bits (redundant bits) are added to the data for error detection.

```
Original Data: 1 0 1 1 0 0 1
                |
Redundant Bits:      0 1 1 0
                |
Encoded Data:  1 0 1 1 0 0 1 0 1 1 0
```

2.3.2 Parity Check

A parity bit is added to make the total number of 1s either even (even parity) or odd (odd parity).

Even Parity

Number of 1s is **even**. Parity bit = 0 if number of 1s is already even; parity bit = 1 if number of 1s is odd.

```
Example 1: Data = 1011001 (Four 1s)
Data: 1 0 1 1 0 0 1
Number of 1s: 4 (Even)
Parity bit: 0
Encoded: 1 0 1 1 0 0 1 0
```

```
Example 2: Data = 1011011 (Five 1s)
Data: 1 0 1 1 0 1 1
Number of 1s: 5 (Odd)
Parity bit: 1
Encoded: 1 0 1 1 0 1 1 1
```

Odd Parity

Number of 1s is **odd**. Parity bit = 1 if number of 1s is even; parity bit = 0 if number of 1s is odd.

Example: Data = 1011001 (Four 1s)
 Parity bit: 1 (to make odd)
 Encoded: 1 0 1 1 0 0 1 1

Two-Dimensional Parity Check (LRC/VRC)

Uses multiple parity bits arranged in a grid — Longitudinal Redundancy Check (LRC) for rows and Vertical Redundancy Check (VRC) for columns.

Data Matrix (4x4):
 1 0 1 1 | 1 (Row parity - odd)
 0 1 0 1 | 0 (Row parity - even)
 1 1 0 0 | 0 (Row parity - even)
 0 0 1 1 | 0 (Row parity - even)
 -----+-----
 1 0 0 1 | 1 (Column parity)

Advantages: Detects single-bit and burst errors; can correct single-bit errors.

Disadvantage: Uses more redundant bits.

2.3.3 Cyclic Redundancy Check (CRC)

CRC is a powerful error detection technique that uses **polynomial division** (cyclic codes).

CRC Process

1. Data is appended with (n-1) zeros where n = degree of generator polynomial
2. Binary division (XOR) of data + zeros by the generator polynomial
3. Remainder is the CRC
4. Codeword = Data + CRC remainder
5. At receiver, divide received data by same polynomial; zero remainder = no error

CRC Example

Problem: Data = 10101, Divisor = 1101
 Step 1: Data = 10101, Divisor = 1101
 Step 2: Append (n-1) = 3 zeros → 10101000
 Step 3: Binary division (XOR):

```

    1101 ) 10101000
           1101
           ----
             1111
             1101
             ----
               110
  
```

```

      ----
      1000
      1101
      ----
      1010
      1101
      ----
      111 (Remainder)

```

Step 4: Remainder = 111

Step 5: Codeword = 10101111

Step 6: Transmit 10101111

At Receiver: Divide 10101111 by 1101 → Remainder = 0 → No error

Polynomial Representation

Term	Example
Divisor	$1101 \rightarrow x^3 + x^2 + 1$
Data	$10101 \rightarrow x^4 + x^2 + 1$
CRC	$x^3 + x^2 + x + 1$

Common CRC Polynomials

Name	Polynomial	Application
CRC-12	$x^{12} + x^{11} + x^3 + x^2 + x + 1$	6-bit characters
CRC-16	$x^{16} + x^{15} + x^2 + 1$	8-bit characters
CRC-CCITT	$x^{16} + x^{12} + x^5 + 1$	8-bit characters
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	Ethernet, ZIP

Advantages: Detects all single-bit, double-bit, odd number of errors, and all burst errors up to polynomial degree.

2.3.4 Checksum

Checksum is an error detection method where the sender adds all data units and sends the sum (checksum) along with the data.

Process

Sender Side: Divide data into fixed-size units (e.g., 16-bit), add using 1's complement, checksum = 1's complement of sum, append to data.

Receiver Side: Add all units including checksum; all 1s = no error; presence of 0s = error.

```
Example: Data = 10101001 00111001
Data:      10101001
           00111001
Sum:       11100010
Checksum:  00011101 (1's complement of sum)
Transmitted: 10101001 00111001 00011101
```

```
Receiver Verification:
10101001
00111001
00011101
-----
11111111 (All 1s → No error)
```

Comparison: Checksum vs CRC

Feature	Checksum	CRC
Method	Addition	Division
Speed	Fast	Slower
Detection Power	Good	Excellent
Implementation	Simple	Complex
Reliability	Less reliable	Very reliable

2.4 Error Correction

Error correction is the process of detecting and **correcting** errors without retransmission.

2.4.1 Forward Error Correction (FEC)

FEC adds redundant data so that errors can be corrected at the receiver without retransmission.

Types: Block Codes (data into blocks) and Convolutional Codes (continuous stream).

Advantages: No retransmission needed, works in one-way communication, real-time correction.

Disadvantages: Overhead (extra bits), more processing power needed.

2.4.2 Automatic Repeat reQuest (ARQ)

ARQ uses acknowledgments and timeouts. If the sender does not receive an ACK within a timeout period, it retransmits the data. Types: Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ.

2.4.3 Hamming Code

Hamming code is a linear error-correcting code that can detect and correct **single-bit errors**.

Formula

Number of parity bits (r) for data bits (d): $2^r \geq d + r + 1$

Parity Bit Positions

Bit Position	Checks Bits
$2^0 = 1$	1, 3, 5, 7, ...
$2^1 = 2$	2, 3, 6, 7, ...
$2^2 = 4$	4, 5, 6, 7, ...
$2^3 = 8$	8, 9, 10, 11, ...

Step-by-Step Example: Encode data 1011

Step 1: $d = 4$, $2^r \geq 4 + r + 1 \rightarrow r = 3$ ($2^3 = 8 \geq 4+3+1 = 8$)

Step 2: 7-bit positions (1-indexed):

Position: 1 2 3 4 5 6 7

Type: P1 P2 D1 P3 D2 D3 D4

Step 3: Place data bits (1011) at positions 3,5,6,7:

Position: 1 2 3 4 5 6 7

Bit: P1 P2 1 P3 0 1 1

Step 4: Calculate parity bits (even parity):

P1 checks positions 1,3,5,7: $?, 1, 0, 1 \rightarrow$ need even $\rightarrow P1 = 0$

P2 checks positions 2,3,6,7: $?, 1, 1, 1 \rightarrow$ need even $\rightarrow P2 = 1$

P3 checks positions 4,5,6,7: $?, 0, 1, 1 \rightarrow$ need even $\rightarrow P3 = 0$

Step 5: Final Hamming code: 0 1 1 0 0 1 1 = 0110011

Error Detection and Correction

Received code: 0110011 (assuming bit 3 flipped $\rightarrow$ 0100011)

P1 checks: 1,3,5,7 = 0,0,0,1 = 1 (odd) $\rightarrow$ Error ($P1 = 1$)

P2 checks: 2,3,6,7 = 1,0,1,1 = 1 (odd) $\rightarrow$ Error ($P2 = 1$)

P3 checks: 4,5,6,7 = 0,0,1,1 = 0 (even) $\rightarrow$ OK ($P3 = 0$)

Error position = $P3 \ P2 \ P1 = 011 = 3$ (bit 3 is in error)

Flip bit 3: $0 \rightarrow 1 \rightarrow$ Corrected code: 0110011 (Data = 1011)

Hamming Code Summary

Feature	Value
Detects	Single and double-bit errors
Corrects	Single-bit errors
Redundancy	r bits for d data bits
Minimum Distance	3

2.4.4 Burst Error Correction (Interleaving)

Original 12 bits (3 words of 4 bits):

Word 1: 1 0 1 1

Word 2: 0 1 0 1

Word 3: 1 0 0 1

Interleaved (column by column): 1 0 1 0 1 0 1 0 0 1 1 1

If burst error affects positions 2-5:

Received: 1 1 1 1 1 0 1 0 0 1 1 1

De-interleave:

Word 1: 1 1 1 1 (1 bit error)

Word 2: 1 1 0 1 (1 bit error)

Word 3: 0 1 1 1 (1 bit error)

Errors spread across words → each word has only 1 error → easier to correct

2.5 Data Communication Media

Transmission Media is the physical path through which data travels from sender to receiver.

- **Guided (Conductive) Media:** Twisted pair, Coaxial, Fiber optic
- **Unguided (Wireless) Media:** Radio waves, Microwaves, Infrared, Satellite

2.6 Conductive Media (Guided Media)

2.6.1 Twisted Pair Cable

Two insulated copper wires twisted together to reduce electromagnetic interference.

Type	Description	Use
UTP (Unshielded)	No shielding	Most common, LAN
STP (Shielded)	Metal shield	High-speed networks
FTP (Foiled)	Foil shield	EMI environments

UTP Categories

Category	Speed	Use
Cat 3	10 Mbps	Voice, 10Base-T
Cat 5	100 Mbps	Fast Ethernet
Cat 5e	1000 Mbps	Gigabit Ethernet
Cat 6	10 Gbps	High-speed networks

Advantages: Inexpensive, easy to install, flexible, widely available.

Disadvantages: Limited distance (100m), susceptible to interference, lower speed than fiber.

2.6.2 Coaxial Cable

Central conductor surrounded by insulation, metal shield, and outer jacket.

Type	Impedance	Use
RG-58	50Ω	Ethernet (ThinNet)
RG-11	75Ω	Cable TV
RG-59	75Ω	Video transmission

Advantages: Better shielding than twisted pair, longer distance (up to 500m), higher bandwidth.

Disadvantages: More expensive, less flexible, harder to install.

2.6.3 Fiber Optic Cable

Transmits data using light pulses through glass or plastic fibers using total internal reflection.

Types

Feature	Single-Mode	Multi-Mode
Core Size	8-10 μm	50-100 μm
Light Source	Laser	LED
Distance	Long (up to 100 km)	Short (up to 2 km)
Bandwidth	Very high	Lower than single-mode
Cost	Expensive	Less expensive

Multi-Mode Fiber Types

Type	Core Size	Bandwidth	Distance
MMF-1	50 μm	500 MHz-km	550m
MMF-2	62.5 μm	200 MHz-km	275m
MMF-3	50 μm	2000 MHz-km	300m

Advantages: Very high bandwidth, long distance, no interference, secure, lightweight.

Disadvantages: Expensive, hard to install, fragile, needs specialized equipment.

Media Comparison Table

Feature	Twisted Pair	Coaxial	Fiber Optic
Bandwidth	Low-Moderate	Moderate	Very High
Distance	Short (100m)	Medium (500m)	Long (100km)
Cost	Low	Medium	High

Installation	Easy	Moderate	Difficult
Interference	Susceptible	Less susceptible	None
Security	Low	Medium	High

2.7 Wireless Transmission

Wireless transmission uses **electromagnetic waves** to transmit data through the air.

2.7.1 Radio Waves

Feature	Description
Frequency	3 kHz – 300 GHz
Propagation	Omnidirectional
Penetration	Good (through walls)
Use	AM/FM radio, TV, WiFi

Advantages: Omnidirectional (no line of sight), good penetration, inexpensive, long distance.

Disadvantages: Low bandwidth, susceptible to interference, security issues.

2.7.2 Microwaves

Feature	Description
Frequency	1–300 GHz
Propagation	Directional
Penetration	Poor (line of sight)
Use	Telecom, Satellite

Types: Terrestrial Microwave (ground towers, 10-50 km) and Satellite Microwave (global coverage, 35,000 km geostationary).

Advantages: High bandwidth, secure (directional), no cable needed.

Disadvantages: Line of sight needed, affected by weather, expensive equipment.

2.7.3 Infrared

Feature	Description
---------	-------------

Frequency	300 GHz – 400 THz
Propagation	Directional
Penetration	Poor
Distance	Short (10m)
Use	Remote controls, short-range

Advantages: Inexpensive, simple, no interference.

Disadvantages: Line of sight required, short distance, no penetration of walls.

2.7.4 Satellite Communication

Satellites in orbit act as repeaters, receiving signals from earth stations and retransmitting them back. Geostationary (35,786 km), Medium Earth Orbit (10,000-20,000 km), Low Earth Orbit (500-2,000 km). Used for TV broadcast, GPS, internet, weather forecasting.

Wireless Media Comparison

Feature	Radio	Microwave	Infrared
Bandwidth	1-100 Mbps	1-10 Gbps	1-10 Mbps
Distance	Long	10-50 km	10m
Cost	Low	High	Low
Line of Sight	Not needed	Required	Required
Penetration	Good	Poor	None

2.8 Transmission Modes

Transmission mode defines the direction of data flow between two devices.

Mode	Direction	Example	Diagram
Simplex	One direction only	Keyboard to monitor, TV broadcast	→
Half-Duplex	Both directions but one at a time	Walkie-talkie	↔ (one at a time)
Full-Duplex	Both directions simultaneously	Telephone call	↔ (both ways)

2.9 Multiplexing

Multiplexing is a technique that allows multiple signals to be transmitted simultaneously over a single communication channel.

Type	Full Form	Principle	Example
FDM	Frequency Division Multiplexing	Different frequencies for each signal	Radio/TV broadcasting
TDM	Time Division Multiplexing	Time slots allocated to each signal	Telephone networks (T1/E1)
WDM	Wavelength Division Multiplexing	Different wavelengths (colors) of light	Fiber optic communication
CDMA	Code Division Multiple Access	Unique codes assigned to each user	3G cellular networks

2.10 Switching

Switching is the mechanism that establishes a communication path between sender and receiver across a network.

Type	Working	Advantages	Disadvantages
Circuit Switching	Dedicated path established before communication	Guaranteed bandwidth, low latency	Inefficient use of resources
Packet Switching	Data broken into packets, each routed independently	Efficient bandwidth usage, fault-tolerant	Variable delay, packet loss possible
Message Switching	Entire message stored and forwarded at each node	No dedicated path needed, traffic management	High delay, requires storage

2.11 Summary Comparisons

Error Detection Comparison

Method	Description	Detection Power	Cost
Parity Check	Single bit	Low	Low
2D Parity (LRC/VRC)	Grid of bits	Medium	Medium
CRC	Polynomial division	Very High	Medium
Checksum	Addition	Medium	Low

Error Correction Comparison

Method	Corrects	Detects	Overhead
Hamming Code	Single-bit	Single/Double	Medium
FEC	Multiple	Multiple	High
Interleaving	Burst errors	Burst errors	High

All Media Comparison

Media	Bandwidth	Distance	Cost	Interference
Twisted Pair	10-1000 Mbps	100m	Low	High
Coaxial	10-1000 Mbps	500m	Medium	Medium
Fiber Optic	1-100 Gbps	100km	High	None
Radio	1-100 Mbps	Long	Low	High
Microwave	1-10 Gbps	10-50km	High	Medium
Infrared	1-10 Mbps	10m	Low	Low

Questions

Short Questions (2 marks)

- What is the difference between single-bit and burst error?
- Define parity check.
- What is CRC?
- What is the difference between error detection and error correction?
- Define Hamming code.
- What are the types of errors in data communication?
- What is checksum?
- Differentiate between UTP and STP.

Medium Questions (5 marks)

- Explain single-bit error and burst error with examples.
- Explain CRC with an example.
- What is Hamming code? Explain with an example.
- Compare twisted pair, coaxial, and fiber optic cables.
- Explain wireless transmission methods.

Long Questions (12 marks)

- (a) Explain different types of errors in data communication.
- (a) What is Hamming code? How does it work?
- (a) Explain different types of guided transmission media.

Additional Questions

Compare FDM, TDM, WDM, and CDMA multiplexing techniques.

Explain circuit switching, packet switching, and message switching with their pros and cons.

What are the different transmission modes? Explain with examples.

Compare simplex, half-duplex, and full-duplex transmission modes.

Explain Forward Error Correction with its advantages and disadvantages.

Write a note on Automatic Repeat Request (ARQ).

What is interleaving? How does it help in burst error correction?

Describe the working of satellite communication.